

[illegible]

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=== Insight 1: Avg Close Price by Day of Week ===
>>> for line in sorted(avg_close.collect(), key=lambda x: -x[1]):
...     print(line)
...
...
```

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at org.apache.spark.api.python.BasePythonRunner$ReaderIterator.handlePythonException(PythonRunner.scala:645)
at org.apache.spark.api.python.PythonRunner$$anon$3.read(PythonRunner.scala:1029)
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at org.apache.spark.util.Utils$.tryWithSafeFinally(Utils.scala:97)

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    at org.apache.spark.executor.Executor$TaskRunner.run(Executor.scala:719)
    at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1136)
    at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:635)
    at java.base/java.lang.Thread.run(Thread.java:840)
26/04/27 20:45:22 ERROR Executor: Exception in task 0.0 in stage 4.0 (TID 6)
org.apache.spark.api.python.PythonException: Traceback (most recent call last):
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/worker.py", line 3386, in main
    process()
    ~~~~~^
File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/worker.py", line 3375, in process
  out_iter = func(split_index, iterator)
File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 5306, in pipeline_func
  return func(split, prev_func(split, iterator))
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File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 5306, in pipeline_func
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  return f(*args, **kwargs)
File "<python-input-11>", line 2, in <lambda>
File "/opt/anaconda3/lib/python3.13/_strptime.py", line 789, in _strptime_datetime
  tt, fraction, gmtoff_fraction = _strptime(data_string, format)
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    at java.base/java.lang.Thread.run(Thread.java:840)
26/04/27 20:45:22 WARN TaskSetManager: Lost task 0.0 in stage 5.0 (TID 8) (192.168.6.13 executor driver): org.apache.spark.api.python.PythonException: Traceback
(most recent call last):
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/worker.py", line 3386, in main
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26/04/27 20:45:22 ERROR TaskSetManager: Task 0 in stage 5.0 failed 1 times; aborting job
26/04/27 20:45:22 WARN TaskSetManager: Lost task 0.0 in stage 4.0 (TID 6) (192.168.6.13 executor driver): org.apache.spark.api.python.PythonException: Traceback
(most recent call last):
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/worker.py", line 3386, in main
    process()
    ~~~~~^
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/worker.py", line 3375, in process
    out_iter = func(split_index, iterator)
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    merger.mergeValues(iterator)
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  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/shuffle.py", line 256, in mergeValues
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26/04/27 20:45:22 ERROR TaskSetManager: Task 0 in stage 4.0 failed 1 times; aborting job
26/04/27 20:45:22 WARN TaskSetManager: Lost task 1.0 in stage 5.0 (TID 9) (192.168.6.13 executor driver): TaskKilled (Stage cancelled: Job aborted due to stage f
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Driver stacktrace:)
26/04/27 20:45:22 WARN TaskSetManager: Lost task 1.0 in stage 4.0 (TID 7) (192.168.6.13 executor driver): TaskKilled (Stage cancelled: Job aborted due to stage failure: Task 0 in stage 5.0 failed 1 times, most recent failure: Lost task 0.0 in stage 5.0 (TID 8) (192.168.6.13 executor driver): org.apache.spark.api.python.PythonException: Traceback (most recent call last):
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Driver stacktrace:)
Traceback (most recent call last):
  File "<python-input-16>", line 1, in <module>
    for line in sorted(avg_close.collect(), key=lambda x: -x[1]):
    ~~~~~^
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 1700, in collect
    sock_info = self.ctx._jvm.PythonRDD.collectAndServe(self._jrdm.rdd())
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/py4j-0.10.9.9-src.zip/py4j/java_gateway.py", line 1362, in __call__
    return_value = get_return_value(
      answer, self.gateway_client, self.target_id, self.name)
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/errors/exceptions/captured.py", line 263, in deco
    return f(*a, **kw)
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/py4j-0.10.9.9-src.zip/py4j/protocol.py", line 327, in get_return_value
    raise Py4JJavaError(
      "An error occurred while calling {0}{1}{2}.\n".
      format(target_id, ".", name), value)
py4j.protocol.Py4JJavaError: An error occurred while calling z:org.apache.spark.api.python.PythonRDD.collectAndServe.
: org.apache.spark.SparkException: Job aborted due to stage failure: Task 0 in stage 5.0 failed 1 times, most recent failure: Lost task 0.0 in stage 5.0 (TID 8) (192.168.6.13 executor driver): org.apache.spark.api.python.PythonException: Traceback (most recent call last):
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/worker.py", line 3386, in main
    process()
    ~~~~~^
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/worker.py", line 3375, in process
    out_iter = func(split_index, iterator)
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 5306, in pipeline_func
    return func(split, prev_func(split, iterator))
    ~~~~~^
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 5306, in pipeline_func
    return func(split, prev_func(split, iterator))
    ~~~~~^
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 705, in func
    return f(iterator)
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 3853, in combineLocally
    merger.mergeValues(iterator)
    ~~~~~^
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/shuffle.py", line 256, in mergeValues
    for k, v in iterator:
    ~~~~~^
  File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/util.py", line 141, in wrapper
    return f(*args, **kwargs)
  File "<python-input-13>", line 2, in <lambda>
  File "/opt/anaconda3/lib/python3.13/_strptime.py", line 789, in _strptime_datetime
    tt, fraction, gmtoff_fraction = _strptime(data_string, format)
    ~~~~~^
  File "/opt/anaconda3/lib/python3.13/_strptime.py", line 555, in _strptime
    raise ValueError("time data %r does not match format %r" %
      (data_string, format))
ValueError: time data 'Date' does not match format '%Y-%m-%d'

    at org.apache.spark.api.python.BasePythonRunner$ReaderIterator.handlePythonException(PythonRunner.scala:645)
    at org.apache.spark.api.python.PythonRunner$$anon$3.read(PythonRunner.scala:1029)
    at org.apache.spark.api.python.PythonRunner$$anon$3.read(PythonRunner.scala:1014)
    at org.apache.spark.api.python.BasePythonRunner$ReaderIterator.hasNext(PythonRunner.scala:596)
    at org.apache.spark.InterruptibleIterator.hasNext(InterruptibleIterator.scala:37)
    at scala.collection.Iterator$$GroupedIterator.fill(Iterator.scala:263)
    at scala.collection.Iterator$$GroupedIterator.hasNext(Iterator.scala:265)

```

```

at scala.collection.Iterator$$anon$9.hasNext(Iterator.scala:593)
at org.apache.spark.shuffle.sort.BypassMergeSortShuffleWriter.write(BypassMergeSortShuffleWriter.java:153)
at org.apache.spark.shuffle.ShuffleWriteProcessor.write(ShuffleWriteProcessor.scala:57)
at org.apache.spark.scheduler.ShuffleMapTask.runTask(ShuffleMapTask.scala:111)
at org.apache.spark.scheduler.ShuffleMapTask.runTask(ShuffleMapTask.scala:54)
at org.apache.spark.TaskContext.runTaskWithListeners(TaskContext.scala:188)
at org.apache.spark.scheduler.Task.run(Task.scala:147)
at org.apache.spark.executor.Executor$TaskRunner.$anonfun$run$5(Executor.scala:716)
at org.apache.spark.util.SparkErrorUtils.tryWithSafeFinally(SparkErrorUtils.scala:86)
at org.apache.spark.util.SparkErrorUtils.tryWithSafeFinally$(SparkErrorUtils.scala:83)
at org.apache.spark.util.Utils$.tryWithSafeFinally(Utils.scala:97)
at org.apache.spark.executor.Executor$TaskRunner.run(Executor.scala:719)
at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1136)
at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:635)
at java.base/java.lang.Thread.run(Thread.java:840)

```

Driver stacktrace:

```

at org.apache.spark.scheduler.DAGScheduler.$anonfun$abortStage$3(DAGScheduler.scala:3122)
at scala.Option.getOrElse(Option.scala:201)
at org.apache.spark.scheduler.DAGScheduler.$anonfun$abortStage$2(DAGScheduler.scala:3122)
at org.apache.spark.scheduler.DAGScheduler.$anonfun$abortStage$2$adapted(DAGScheduler.scala:3114)
at scala.collection.immutable.List.foreach(List.scala:323)
at org.apache.spark.scheduler.DAGScheduler.abortStage(DAGScheduler.scala:3114)
at org.apache.spark.scheduler.DAGScheduler.$anonfun$handleTaskSetFailed$1(DAGScheduler.scala:1303)
at org.apache.spark.scheduler.DAGScheduler.$anonfun$handleTaskSetFailed$1$adapted(DAGScheduler.scala:1303)
at scala.Option.foreach(Option.scala:437)
at org.apache.spark.scheduler.DAGScheduler.handleTaskSetFailed(DAGScheduler.scala:1303)
at org.apache.spark.scheduler.DAGSchedulerEventProcessLoop.doOnReceive(DAGScheduler.scala:3397)
at org.apache.spark.scheduler.DAGSchedulerEventProcessLoop.onReceive(DAGScheduler.scala:3328)
at org.apache.spark.scheduler.DAGSchedulerEventProcessLoop.onReceive(DAGScheduler.scala:3317)
at org.apache.spark.util.EventLoop$$anon$1.run(EventLoop.scala:50)
at org.apache.spark.scheduler.DAGScheduler.runJob(DAGScheduler.scala:1017)
at org.apache.spark.SparkContext.runJob(SparkContext.scala:2496)
at org.apache.spark.SparkContext.runJob(SparkContext.scala:2517)
at org.apache.spark.SparkContext.runJob(SparkContext.scala:2536)
at org.apache.spark.SparkContext.runJob(SparkContext.scala:2561)
at org.apache.spark.rdd.RDD.$anonfun$collect$1(RDD.scala:1057)
at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:151)
at org.apache.spark.rdd.RDDOperationScope$.withScope(RDDOperationScope.scala:112)
at org.apache.spark.rdd.RDD.withScope(RDD.scala:417)
at org.apache.spark.rdd.RDD.collect(RDD.scala:1056)
at org.apache.spark.api.python.PythonRDD$.collectAndServe(PythonRDD.scala:205)
at org.apache.spark.api.python.PythonRDD.collectAndServe(PythonRDD.scala)
at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(Native Method)
at java.base/jdk.internal.reflect.NativeMethodAccessorImpl.invoke(NativeMethodAccessorImpl.java:77)
at java.base/jdk.internal.reflect.DelegatingMethodAccessorImpl.invoke(DelegatingMethodAccessorImpl.java:43)
at java.base/java.lang.reflect.Method.invoke(Method.java:569)
at py4j.reflection.MethodInvoker.invoke(MethodInvoker.java:244)
at py4j.reflection.ReflectionEngine.invoke(ReflectionEngine.java:374)
at py4j.Gateway.invoke(Gateway.java:282)
at py4j.commands.AbstractCommand.invokeMethod(AbstractCommand.java:132)
at py4j.commands.CallCommand.execute(CallCommand.java:79)
at py4j.ClientServerConnection.waitForCommands(ClientServerConnection.java:184)
at py4j.ClientServerConnection.run(ClientServerConnection.java:108)
at java.base/java.lang.Thread.run(Thread.java:840)

```

Caused by: org.apache.spark.api.python.PythonException: Traceback (most recent call last):

```

File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/worker.py", line 3386, in main
process()
~~~~~
File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/worker.py", line 3375, in process
out_iter = func(split_index, iterator)
File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 5306, in pipeline_func
return func(split, prev_func(split, iterator))
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File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 5306, in pipeline_func
return func(split, prev_func(split, iterator))
File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 705, in func
return f(iterator)
File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/pyspark/core/rdd.py", line 3853, in combineLocally
merger.mergeValues(iterator)
~~~~~
File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/shuffle.py", line 256, in mergeValues
for k, v in iterator:
~~~~~
File "/Users/gajendra06/Documents/dev/apache/spark-4.1.1-bin-hadoop3/python/lib/pyspark.zip/pyspark/util.py", line 141, in wrapper
return f(*args, **kwargs)
File "<python-input-13>", line 2, in <lambda>
File "/opt/anaconda3/lib/python3.13/_strptime.py", line 789, in _strptime_datetime
tt, fraction, gmtoff_fraction = _strptime(data_string, format)
~~~~~
File "/opt/anaconda3/lib/python3.13/_strptime.py", line 555, in _strptime
raise ValueError("time data %r does not match format %r" %
(data_string, format))
ValueError: time data 'Date' does not match format '%Y-%m-%d'

```

```

at org.apache.spark.api.python.BasePythonRunner$ReaderIterator.handlePythonException(PythonRunner.scala:645)
at org.apache.spark.api.python.PythonRunner$$anon$3.read(PythonRunner.scala:1029)
at org.apache.spark.api.python.PythonRunner$$anon$3.read(PythonRunner.scala:1014)
at org.apache.spark.api.python.BasePythonRunner$ReaderIterator.hasNext(PythonRunner.scala:596)
at org.apache.spark.InterruptibleIterator.hasNext(InterruptibleIterator.scala:37)
at scala.collection.Iterator$GroupedIterator.fill(Iterator.scala:263)
at scala.collection.Iterator$GroupedIterator.hasNext(Iterator.scala:265)
at scala.collection.Iterator$$anon$9.hasNext(Iterator.scala:593)
at org.apache.spark.shuffle.sort.BypassMergeSortShuffleWriter.write(BypassMergeSortShuffleWriter.java:153)
at org.apache.spark.shuffle.ShuffleWriteProcessor.write(ShuffleWriteProcessor.scala:57)
at org.apache.spark.scheduler.ShuffleMapTask.runTask(ShuffleMapTask.scala:111)
at org.apache.spark.scheduler.ShuffleMapTask.runTask(ShuffleMapTask.scala:54)
at org.apache.spark.TaskContext.runTaskWithListeners(TaskContext.scala:188)
at org.apache.spark.scheduler.Task.run(Task.scala:147)
at org.apache.spark.executor.Executor$TaskRunner.$anonfun$run$5(Executor.scala:716)
at org.apache.spark.util.SparkErrorUtils.tryWithSafeFinally(SparkErrorUtils.scala:86)
at org.apache.spark.util.SparkErrorUtils.tryWithSafeFinally$(SparkErrorUtils.scala:83)
at org.apache.spark.util.Utils$.tryWithSafeFinally(Utils.scala:97)
at org.apache.spark.executor.Executor$TaskRunner.run(Executor.scala:719)
at java.base/java.util.concurrent.ThreadPoolExecutor.runWorker(ThreadPoolExecutor.java:1136)
at java.base/java.util.concurrent.ThreadPoolExecutor$Worker.run(ThreadPoolExecutor.java:635)
... 1 more

```

```

>>> rdd1 = sc.textFile("/Users/gajendra06/Documents/HCLTECH.csv")
>>> header = rdd1.first()
>>> rdd2 = rdd1.filter(lambda a: a != header) \

```

```

...         .filter(lambda a: not a.startswith("Ticker")) \
...         .filter(lambda a: not a.startswith("Price")) \
...         .filter(lambda a: not a.startswith("Date"))
>>>
>>> print("Total rows:", rdd2.count())
Total rows: 5529
>>> for line in rdd2.take(3):
...     print(line)
...
2002-08-12 00:00:00+00:00,13.796875,23.762500762939453,23.987499237060547,22.625,22.712499618530273,11239400
2002-08-13 00:00:00+00:00,13.542858123779297,23.325000762939453,24.25,23.149999618530273,24.25,3756368
2002-08-14 00:00:00+00:00,13.357786178588867,23.006250381469727,23.237499237060547,22.768749237060547,23.237499237060547,3964016
>>> from datetime import datetime
... format_date = "%Y-%m-%d"
...
>>> rdd3 = rdd2.map(lambda a: (
...     datetime.strptime(a.split(",")[0][:10], format_date).strftime("%A"),
...     float(a.split(",")[2])
... ))
>>> rdd4 = rdd3.reduceByKey(lambda a, b: a + b)
>>> rdd_cnt = rdd2.map(lambda a: (
...     datetime.strptime(a.split(",")[0][:10], format_date).strftime("%A"), 1
... )).reduceByKey(lambda a, b: a + b)
>>> avg_close = rdd4.join(rdd_cnt).map(lambda a: (a[0], round(a[1][0]/a[1][1], 2)))
>>> print("\n=== Insight 1: Avg Close Price by Day of Week ===")
... for line in sorted(avg_close.collect(), key=lambda x: -x[1]):
...     print(line)
...
=== Insight 1: Avg Close Price by Day of Week ===
('Sunday', 565.25)
('Saturday', 457.68)
('Thursday', 402.31)
('Wednesday', 401.91)
('Tuesday', 401.47)
('Friday', 399.48)
('Monday', 399.04)
>>> rdd_vol = rdd2.map(lambda a: (
...     datetime.strptime(a.split(",")[0][:10], format_date).strftime("%B"),
...     float(a.split(",")[6])
... ))
>>> vol_sum = rdd_vol.reduceByKey(lambda a, b: a + b)
>>> vol_cnt = rdd2.map(lambda a: (
...     datetime.strptime(a.split(",")[0][:10], format_date).strftime("%B"), 1
... )).reduceByKey(lambda a, b: a + b)
>>> avg_vol = vol_sum.join(vol_cnt).map(lambda a: (a[0], round(a[1][0]/a[1][1], 0)))
>>> print("\n=== Insight 2: Avg Volume by Month ===")
... for line in sorted(avg_vol.collect(), key=lambda x: -x[1]):
...     print(line)
...
=== Insight 2: Avg Volume by Month ===
('October', 6634670.0)
('April', 5631844.0)
('September', 5448835.0)
('January', 4939978.0)
('November', 4913967.0)
('December', 4778224.0)
('March', 4719560.0)
('May', 4621297.0)
('August', 4314229.0)
('July', 4212827.0)
('June', 4091963.0)
('February', 4074477.0)
>>> rdd_range = rdd2.map(lambda a: (
...     a.split(",")[0][:10],
...     round(float(a.split(",")[3]) - float(a.split(",")[4]), 2) # High - Low
... ))
...
... print("\n=== Insight 3: Top 10 Most Volatile Days ===")
... for line in rdd_range.sortBy(lambda a: -a[1]).take(10):
...     print(line)
...
=== Insight 3: Top 10 Most Volatile Days ===
('2020-03-13', 121.3)
('2024-06-04', 90.0)
('2021-10-11', 89.25)
('2024-10-31', 87.35)
('2020-09-14', 80.95)
('2021-09-27', 77.9)
('2023-12-15', 75.0)
('2024-03-14', 73.0)
('2021-10-12', 72.7)
('2024-10-23', 71.5)
>>> rdd_year = rdd2.map(lambda a: (
...     a.split(",")[0][:4],
...     float(a.split(",")[2])
... ))
>>> yr_sum = rdd_year.reduceByKey(lambda a, b: a + b)
... yr_cnt = rdd2.map(lambda a: (
...     a.split(",")[0][:4], 1
... )).reduceByKey(lambda a, b: a + b)
...
>>> avg_yr = yr_sum.join(yr_cnt) \
...     .map(lambda a: (a[0], round(a[1][0]/a[1][1], 2))) \
...     .sortBy(lambda a: a[0])
>>> print("\n=== Insight 4: Year-wise Avg Close Price ===")
... for line in avg_yr.collect():
...     print(line)
...
=== Insight 4: Year-wise Avg Close Price ===
('2002', 23.89)
('2003', 22.58)
('2004', 39.77)
('2005', 50.64)
('2006', 71.92)
('2007', 78.88)
('2008', 57.83)
('2009', 54.39)

```

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('2010', 97.27)
('2011', 113.74)
('2012', 130.86)
('2013', 220.4)
('2014', 374.96)
('2015', 458.75)
('2016', 396.03)
('2017', 429.4)
('2018', 488.6)
('2019', 534.71)
('2020', 661.43)
('2021', 1061.48)
('2022', 1055.42)
('2023', 1176.93)
('2024', 1597.73)
>>> rdd_bull = rdd2.map(lambda a: (
...     "Bullish" if float(a.split(",")[2]) > float(a.split(",")[5]) else "Bearish", 1
... ))
>>> bull_count = rdd_bull.reduceByKey(lambda a, b: a + b)
>>> print("\n=== Insight 5: Bullish vs Bearish Days ===")
... for line in bull_count.collect():
...     print(line)
...

=== Insight 5: Bullish vs Bearish Days ===
('Bullish', 2607)
('Bearish', 2922)
>>>

```